Adapting the Chatterbox Text-to-Speech Architecture to Romanian via Phoneme-Level Preprocessing

Adrian Stanea
[Institution]
{email}

Abstract*—Romanian remains unsupported in most state-of-the-art (SOTA) zero-shot text-to-speech (TTS) systems, including Chatterbox — a 0.5-billion-parameter, MIT-licensed model based on a Llama transformer backbone with conditional flow matching (CFM). We present the first adaptation of the Chatterbox architecture to a new language, using Romanian as the target. Our approach fine-tunes only the T3 text-to-token transformer on the SWARA 1.0 corpus (21,304 utterances, ∼21 hours, 18 speakers), while leaving the voice encoder and S3 speech generator frozen. A central contribution is a phoneme-level text preprocessing strategy that maps the two Romanian characters absent from the existing 2,454-token grapheme vocabulary (ş → *sh*, ţ → *ts*) to phonetically equivalent existing tokens, fully preserving pretrained embedding quality. We demonstrate that the alternative approach — extending the vocabulary with mean-initialized embeddings — induces posterior collapse, a failure mode independently reported across multiple languages and corroborated here experimentally. After 13,000 training steps (∼39 epochs) with a learning rate of 5×10^{-6}, the model produces intelligible Romanian speech from unseen reference speakers. Systematic ablation over exaggeration and classifier-free guidance (CFG) inference parameters on two zero-shot holdout speakers reveals a consistent trade-off: low exaggeration favours intelligibility (best CER 8.7%, WER 25.9%, UTMOS 3.46) while higher exaggeration improves speaker fidelity (cosine similarity up to 0.875). These results establish phoneme-level preprocessing as a viable, architecture-neutral strategy for adapting BPE-tokenised TTS models to new languages without architectural modification.

***Index Terms*—text-to-speech synthesis, language adaptation, Romanian, flow matching, zero-shot voice cloning, fine-tuning, posterior collapse**

I. Introduction and Related Work

A. Introduction

The past three years have fundamentally altered the landscape of neural TTS. VALL-E [1] reframed speech synthesis as conditional language modelling over discrete audio codec tokens, demonstrating zero-shot voice cloning from three seconds of reference audio. Concurrent and subsequent work explored flow matching as an alternative generative backbone: rather than iteratively denoising a signal, CFM learns a vector field that transports noise samples to speech along a straight optimal-transport path defined by an ordinary differential equation (ODE), enabling stable, high-fidelity synthesis with far fewer inference steps [3]. CosyVoice [4] combined supervised semantic tokens derived from a multilingual ASR model with chunk-aware causal flow matching, achieving human-parity naturalness across nine languages.

Despite this progress, Romanian remains a low-resource language in the TTS domain. Existing Romanian synthesis options span three regimes: (i) MMS-TTS-RON [8], a single-speaker VITS model derived from religious-text data, covering the language but offering no voice cloning capability and below-SOTA quality; (ii) FastPitch-RO [6], a non-autoregressive duration-model system adapted to 18 speakers via TitaNet embeddings on SWARA, offering multi-speaker synthesis but without zero-shot generalisation; (iii) F5-TTS-RO [7], the most recent and most directly relevant work, which introduces a lightweight ConvNeXt input adapter that keeps the F5-TTS backbone frozen, achieving WER 5.27% and speaker cosine similarity 0.90 on Common Voice — though residual English accent artefacts were noted in a subset of samples.

Chatterbox [5] is a family of SOTA open-source TTS models (MIT license) that has not been adapted to Romanian. Its original variant employs a 0.5B-parameter Llama backbone for flow matching, trained on approximately 500,000 hours of cleaned speech data, and provides zero-shot voice cloning with a unique emotion exaggeration control parameter. Notably, the Chatterbox architecture differs substantively from F5-TTS: it uses a BPE grapheme tokeniser with 2,454 tokens (not character padding), and processes speech through a two-stage T3/S3 pipeline rather than a single DiT backbone. Adapting Chatterbox thus poses distinct challenges — and this work constitutes the first published adaptation of Chatterbox to any new language.

Our contributions are:

1) A phoneme-level preprocessing strategy for adapting BPE-tokenised TTS models to new languages without vocabulary extension, demonstrated to prevent posterior collapse.
2) Fine-tuning of Chatterbox Multilingual on Romanian SWARA data, producing intelligible zero-shot synthesis.
3) Systematic ablation over exaggeration and CFG inference parameters, characterising the intelligibility–speaker-fidelity trade-off.

B. Related Work

1) Modern Zero-Shot TTS Architectures: The neural codec language model paradigm, introduced by VALL-E [1], established that discrete audio tokens produced by a neural

codec (EnCodec) can serve as the prediction target for a language model conditioned on a short acoustic prompt. Seed-TTS [2] extended this with reinforcement learning for improved robustness and speaker similarity. F5-TTS [3] took a non-autoregressive approach: a Diffusion Transformer (DiT) trained with flow matching, requiring no duration model or explicit phoneme alignments, and demonstrating strong code-switching on a 100k-hour multilingual corpus. CosyVoice [4] introduced supervised semantic tokens — derived from a multilingual ASR encoder — as a bridge between text and flow-matching-based speech generation, with its successor CosyVoice 2 achieving human-parity streaming synthesis.

2) Chatterbox Architecture Family: Chatterbox [5] is built around a 0.5B-parameter Llama 3 transformer that maps text tokens to speech tokens (T3), followed by a frozen S3 generator that synthesises mel-spectrograms via CFM. All audio includes a PerTh perceptual watermark. The Multilingual variant supports 23 languages (excluding Romanian) with a 2,454-token grapheme tokeniser. The Turbo variant (350M parameters, GPT-2 backbone, single-step decoding) has documented instability on new languages due to its BPE tokeniser and has been excluded from consideration here.

3) Romanian TTS and Low-Resource Adaptation: FastPitch-RO [6] adapted the FastPitch architecture to 18 Romanian speakers by conditioning on TitaNet-L speaker embeddings extracted from SWARA. F5-TTS-RO [7] demonstrated that a lightweight learnable input adapter (ConvNeXt) could extend F5-TTS to Romanian while keeping the entire model frozen, with a key finding that full fine-tuning yielded better WER (3.62%) but substantially lower speaker similarity (0.79) compared to the adapter (5.27%, 0.90), highlighting the inherent trade-off in low-resource adaptation. MMS-TTS [8] scaled VITS-based synthesis to 1,100 languages including Romanian, but operates as a single-speaker system. Align2Speak [9] proposed a GRPO-based framework that uses ASR-derived reward signals to align multilingual TTS models to low-resource languages, reporting CER reduction from 33% to 3.94% with only 30 minutes of Portuguese data.

The SWARA corpus [10] provides 21 hours of high-quality Romanian read speech from 17 speakers under CC-BY-SA 4.0, and is the primary training resource for Romanian TTS research.

II. Theoretical Overview

A. Conditional Flow Matching for Speech Synthesis

Flow matching [3] defines a generative model by learning a time-dependent vector field $v_t(x)$ that maps samples from a simple prior distribution p_0 (typically Gaussian noise) to the data distribution p_1 (speech mel-spectrograms) via an ODE:

$$\frac{dx}{dt} = v_t(x), \quad x(0) \sim p_0. \tag{1}$$

Training minimises the conditional flow matching (CFM) objective:

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t,p_1(x_1),p_t(x|x_1)} \left\| v_t(x) - u_t(x \mid x_1) \right\|^2, \tag{2}$$

where $u_t(x \mid x_1)$ is the conditional vector field for the optimal-transport (OT) path between x_0 and x_1:

$$u_t(x \mid x_1) = x_1 - x_0, \quad x_t = (1-t)\, x_0 + t\, x_1. \tag{3}$$

Unlike diffusion models — which require dozens to hundreds of denoising steps — OT-CFM interpolates linearly between noise and data, enabling both faster training convergence and inference in as few as 10–20 ODE steps with a numerical solver. Classifier-free guidance (CFG) scales the conditional signal at inference:

$$\tilde{v}_t = v_t^{\varnothing} + \alpha\big(v_t^{\text{cond}} - v_t^{\varnothing}\big), \tag{4}$$

where α is the CFG scale and $v_t^{\varnothing}$ is the unconditional vector field. Higher α strengthens the conditioning signal at the cost of increased mode-seeking behaviour.

B. The Chatterbox Architecture

Chatterbox processes text-to-speech synthesis in four stages, as illustrated conceptually below.

Stage 1 — EnTokenizer. Input text is converted to a sequence of integer token IDs by a BPE grapheme tokeniser containing 2,454 tokens covering Latin, Cyrillic, and other scripts across 23 languages. Text is lowercased and punctuation-normalised before tokenisation.

Stage 2 — T3 (Text-to-Token Transformer). The T3 model is a 500M-parameter Llama 3-based autoregressive transformer conditioned on (a) the text token sequence and (b) a speaker embedding extracted by the Voice Encoder from a short reference audio clip. It predicts a sequence of discrete speech tokens representing the phonetic content and prosody of the target utterance. This is the *only trainable component* during fine-tuning; all other modules remain frozen.

Stage 3 — S3Gen (Speech-to-Signal Generator). S3Gen synthesises a mel-spectrogram from the speech token sequence via CFM, additionally conditioned on the prompt mel-spectrogram and a speaker embedding. The CFM backbone is an OT-flow-matching model with a Llama transformer as the vector-field estimator. Being frozen, S3Gen retains its full pretrained capacity for high-fidelity acoustic generation.

Stage 4 — Vocoder. A neural vocoder (also frozen) converts the mel-spectrogram to a time-domain waveform at 24 kHz.

Inference parameters. Two parameters are exposed at inference time:

- *Exaggeration* (exag $\in [0, 1]$): scales the emotional intensity of generated speech by modulating the conditioning signal strength in the T3 model.
- *Classifier-free guidance scale* (cfg $\in \mathbb{R}$): the α parameter in Eq. (4), controlling the trade-off between text-fidelity and acoustic diversity.

C. Evaluation Metrics

Intelligibility — CER and WER. Synthesised audio is transcribed using a Romanian ASR model [12]; the transcript is compared against the reference text with standard dynamic-programming alignment. CER operates at the character level; WER at the word level. Both metrics are lower-is-better.

Speech quality — UTMOS. The UTokyo-SaruLab MOS predictor [11] is a neural network trained on the VoiceMOS Challenge 2022 dataset to predict Mean Opinion Score (MOS, 1–5 scale) from a waveform without human listeners. UTMOS generalises reasonably across languages and recording conditions, making it suitable as an automatic quality proxy.

Speaker similarity. Speaker embeddings are extracted from both the synthesised audio and the reference speaker clip using the same encoder used by Chatterbox's Voice Encoder. Cosine similarity between the two embedding vectors quantifies voice-cloning fidelity, ranging from 0 (orthogonal) to 1 (identical).

III. Methodology

A. Dataset: SWARA 1.0

The SWARA Speech Corpus [10] is the primary open resource for Romanian TTS research. We use SWARA 1.0, comprising 21,304 utterances recorded from 43 speaker-session variants (18 unique speakers across multiple recording rounds) at 22,050 Hz. Utterances contain an average of 10.4 words (62.4 characters), with 64 unique characters including all five Romanian diacritic classes. Audio files were converted from the original format to the LJSpeech directory layout (WAV + CSV metadata) for compatibility with the training pipeline.

Dataset splits follow a zero-shot evaluation design: two speakers (BAS and SGS, spanning three recording rounds each — 5 speaker-session variants totalling 2,484 utterances) are held out entirely for evaluation. The remaining utterances are split 90/10 per speaker into training (18,740 samples, 38 speaker-variants) and validation (80 samples, 36 speaker-variants). This strategy ensures that both the acoustic characteristics and the recording-condition variation of the holdout speakers are completely unseen during training, providing a rigorous zero-shot voice-cloning test.

Romanian presents specific phonological challenges for adaptation. The language uses five diacritic characters: ă (central mid-vowel, similar to English schwa), â and î (close central unrounded vowel, a phone absent from most Western European languages), ş (palato-alveolar fricative, as in English "shoe"), and ţ (affricate /ts/, as in English "cats"). Two of these (ş, ţ) are absent from the 23-language grapheme vocabulary, motivating our preprocessing strategy. Contemporary Romanian orthography admits both cedilla variants (ş, ţ) and comma-below variants for the same phones; our preprocessor handles both Unicode forms to ensure full corpus coverage.

B. Language Adaptation: Three Iterations

1) Iteration 1 — Vocabulary Extension (Failed): The community fine-tuning toolkit for Chatterbox recommends extending the tokeniser vocabulary with missing characters for new languages. Following this approach, we added five Romanian tokens (ş, ţ, Ş, Ţ, Ă), expanding the vocabulary from 2,454 to 2,459 tokens. The T3 model's embedding layer and output projection were resized; new token embeddings were initialised with the mean of all existing embeddings (mean initialisation).

Fine-tuning with this configuration caused the model to produce unintelligible speech within a few hundred training steps. Analysis indicates *posterior collapse*: the T3 model ignores the text conditioning path in favour of the stronger audio signal from the voice encoder. Mean-initialised embeddings are non-discriminative — positioned at the centroid of the embedding space, equidistant from all existing token clusters — providing a far weaker conditioning signal than the well-trained audio encoder. The model exploits this imbalance and degenerates into an autoencoder that copies the reference audio regardless of text input. This failure was independently reported by community users of the same toolkit for Turkish, Norwegian, German, and Arabic, and confirmed as a structural problem by the toolkit maintainer.

2) Iteration 2 — Phoneme Mapping with Unstable Hyperparameters: Two key observations motivated an alternative strategy: (i) only two Romanian characters (ş, ţ) are genuinely absent from the 2,454-token vocabulary — the vowels ă, â, î are already present; (ii) both missing characters have phonetically exact equivalents already in the vocabulary: ş ↔ *sh* (IPA /S/) and ţ ↔ *ts* (IPA /ts/). We therefore implemented a phoneme-level preprocessor that replaces the two absent consonants with their existing multi-character equivalents before tokenisation, while leaving all other characters unchanged. This preserves the full 2,454-token vocabulary without any mean-initialised embeddings, retaining the discriminative quality of all pretrained representations.

However, the initial training run at a learning rate that was too high produced unstable outputs — the model generated unintelligible speech after a limited number of training steps, indicating that the T3 transformer is sensitive to learning-rate magnitude during fine-tuning.

3) Iteration 3 — Stabilised Phoneme Mapping (Final Configuration): The phoneme-level preprocessing from Iteration 2 was retained; the primary change was a conservative reduction of the learning rate and an increase in batch size. Table I summarises the three iterations.

TABLE I
Comparison of the three adaptation iterations.

	Iter. 1	**Iter. 2**	**Iter. 3**
Vocabulary size	2,459	2,454	2,454
Preprocessing	None	Phoneme	Phoneme
Embedding init	Mean	Pretrained	Pretrained
Learning rate	high	too high	5×10^{-6}
Batch size	small	small	16
Outcome	Collapse	Unstable	Stable

The phoneme mappings applied in Iterations 2 and 3 are summarised in Table II.

C. Training Setup

Only the T3 transformer is fine-tuned; the Voice Encoder and S3Gen remain frozen throughout. Training was conducted on an NVIDIA V100 (32 GB VRAM) using FP16 precision. Key hyperparameters: learning rate 5×10^{-6}, batch size 16, gradient accumulation 1, maximum text length 256 tokens, maximum speech length 850 tokens, prompt duration 3.0 s.

TABLE II
ROMANIAN PHONEME PREPROCESSING MAPPINGS.

Character	IPA	Mapped to	Token ID
ş, Ş	/sh/	*sh*	120
ţ, Ţ	/ts/	*ts*	192
ă	/ə/	*kept*	2413
â	/1/	*kept*	395
î	/1/	*kept*	407

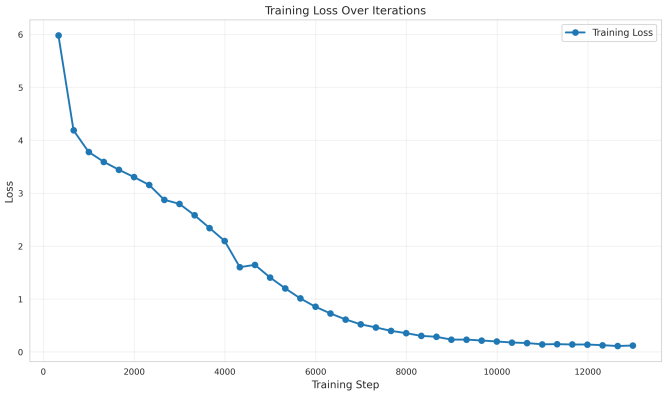


Fig. 1. Training loss over 13,000 steps (≈39 epochs). The loss decreases monotonically from 5.98 to 0.12, with rapid initial descent and gradual plateau after step ∼10,000.

We trained for 13,000 global steps (≈39 epochs), saving checkpoints every 100 steps (retaining the last 10). Training was monitored via TensorBoard; no learning-rate schedule was applied.

D. Evaluation Protocol

Each of the two holdout speakers (BAS, SGS) was evaluated under four parameter combinations: exaggeration $\in \{0.3, 0.7\}$ $\times$ CFG scale $\in \{0.3, 0.5\}$, yielding eight experimental conditions. For each condition, 50 test utterances were synthesised using a 3-second speaker reference clip, producing 400 total samples. CER and WER were computed via transcription with a Romanian ASR model [12]. UTMOS [11] and speaker cosine similarity were computed per sample. All statistics are reported as mean $\pm$ standard deviation over the 50-sample per-condition pool.

IV. Evaluation and Experimental Results

A. Training Dynamics

Fig. 1 shows the training loss curve. The model converges monotonically, with an initial sharp descent from 5.98 to approximately 1.6 within the first 4,000 steps, followed by a gradual refinement phase that reaches a minimum of 0.11 at step 13,000 — a 98% total reduction. The smooth trajectory, with no oscillation or divergence, confirms that the learning-rate reduction implemented in Iteration 3 provides stable training dynamics. The plateau observed after step ∼10,000 suggests the model has captured the core Romanian phonetic patterns within the fixed-vocabulary representation; extended training at a lower learning rate remains a promising avenue for further refinement.

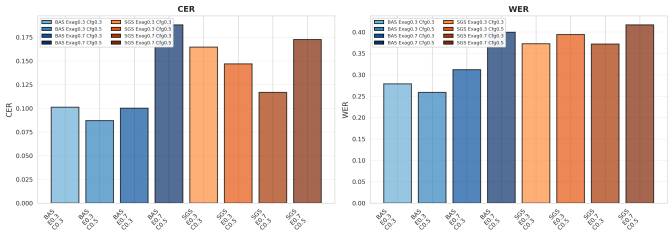


Fig. 2. CER (left) and WER (right) across all eight experimental conditions. BAS speaker (blue) consistently achieves lower error rates than SGS (orange). Low exaggeration (Exag=0.3) generally favours intelligibility.

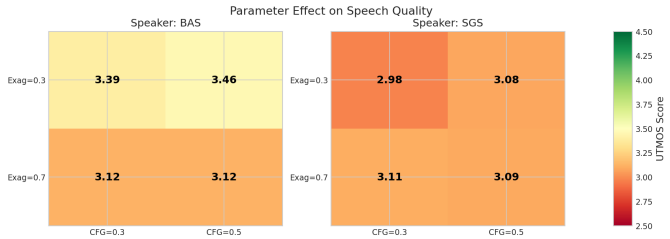


Fig. 3. UTMOS quality scores for BAS (left) and SGS (right) across exaggeration and CFG parameter combinations. BAS at low exaggeration achieves the highest perceived quality (3.39–3.46).

B. Intelligibility

Table III and Fig. 2 present intelligibility results across all conditions. The best intelligibility is achieved by BAS at exag=0.3, cfg=0.5 (CER 8.7%, WER 25.9%). BAS consistently outperforms SGS across all parameter combinations: aggregated mean CER 11.9% vs 15.0%, mean WER 31.2% vs 38.9% (Table IV).

The effect of the exaggeration parameter is clear for BAS: low exaggeration (0.3) yields CER $\approx$ 9–10%, while high exaggeration (0.7) raises CER to $\approx$ 10–19%. The anomalously high variance at BAS exag=0.7, cfg=0.5 (CER std 0.49) indicates occasional catastrophic generation failures under this combination, suggesting that high-exaggeration generation in BAS is more susceptible to inference instability. For SGS, the exaggeration effect on intelligibility is more modest, with CER ranging from 11.7% to 17.3% across conditions.

CFG scale has a secondary but observable effect: for BAS at low exaggeration, cfg=0.5 reduces both CER and WER relative to cfg=0.3. This is consistent with the theoretical expectation that stronger CFG reinforces the text-conditioning signal, improving phoneme reproduction accuracy.

C. Speech Quality

Fig. 3 shows UTMOS quality scores as a heatmap over the parameter grid for each speaker. The best quality is achieved by BAS at exag=0.3, cfg=0.5 (UTMOS 3.46). Low exaggeration consistently yields higher UTMOS for BAS (3.39–3.46 vs 3.12 at exag=0.7), while the effect is less pronounced for

TABLE III
OBJECTIVE EVALUATION METRICS FOR ROMANIAN TTS (MEAN ± STD). BEST INTELLIGIBILITY PER METRIC IS IN **BOLD**.

Condition	CER↓	WER↓	UTMOS↑	Spk. Sim.↑
BAS (exag=0.3, cfg=0.3)	0.101 ± 0.143	0.279 ± 0.233	3.393 ± 0.570	0.810 ± 0.075
BAS (exag=0.3, cfg=0.5)	$\mathbf{0.087 \pm 0.141}$	$\mathbf{0.259 \pm 0.222}$	$\mathbf{3.460 \pm 0.306}$	0.816 ± 0.058
BAS (exag=0.7, cfg=0.3)	0.100 ± 0.116	0.312 ± 0.254	3.123 ± 0.347	0.824 ± 0.054
BAS (exag=0.7, cfg=0.5)	0.188 ± 0.490	0.399 ± 0.484	3.120 ± 0.326	0.830 ± 0.050
SGS (exag=0.3, cfg=0.3)	0.165 ± 0.233	0.373 ± 0.297	2.975 ± 0.504	0.830 ± 0.148
SGS (exag=0.3, cfg=0.5)	0.147 ± 0.153	0.394 ± 0.256	3.082 ± 0.302	0.863 ± 0.084
SGS (exag=0.7, cfg=0.3)	0.117 ± 0.112	0.372 ± 0.291	3.105 ± 0.233	$\mathbf{0.875 \pm 0.040}$
SGS (exag=0.7, cfg=0.5)	0.173 ± 0.193	0.417 ± 0.287	3.094 ± 0.216	0.872 ± 0.053

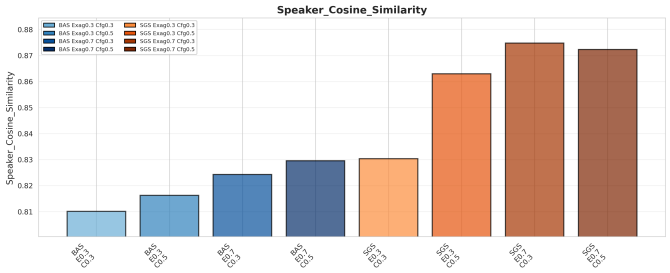


Fig. 4. Speaker cosine similarity across conditions. SGS (orange) consistently achieves higher fidelity than BAS (blue). Higher exaggeration improves speaker preservation for both speakers.

SGS (2.98–3.08 vs 3.09–3.11). BAS achieves higher quality overall (aggregated mean 3.27 vs 3.06 for SGS).

UTMOS scores in the 3.0–3.5 range correspond to "fair to good" perceived quality on the 1–5 scale — confirming intelligible synthesis while indicating headroom for improvement relative to the Chatterbox base model's performance on English. The quality advantage of low exaggeration aligns with the intelligibility findings, suggesting that exaggeration uniformly degrades both dimensions of perceived output quality.

D. Speaker Fidelity

Fig. 4 reveals a pattern complementary to the intelligibility results. SGS consistently achieves higher speaker cosine similarity (0.830–0.875) than BAS (0.810–0.830), inverting the intelligibility ranking. Critically, higher exaggeration improves speaker similarity for both speakers: BAS increases from ≈ 0.813 (exag=0.3) to ≈ 0.827 (exag=0.7); SGS from ≈ 0.847 to ≈ 0.873. This creates a clear parameter-driven trade-off: the settings that maximise intelligibility degrade speaker fidelity, and vice versa.

The positive effect of CFG on speaker similarity is also observable: higher cfg=0.5 slightly outperforms cfg=0.3 at all conditions for SGS, while the effect for BAS is mixed. All eight conditions exceed cosine similarity 0.80, confirming that the frozen Voice Encoder successfully transfers speaker characteristics to unseen Romanian speakers via the fine-tuned T3.

E. Speaker-Level Comparison

Table IV aggregates results across all parameter conditions per speaker. BAS exhibits stronger intelligibility and quality

TABLE IV
SPEAKER-LEVEL METRICS (MEAN ± STD, ACROSS ALL CONDITIONS).

Spk.	CER	WER	UTMOS	Sim.
BAS	0.119±0.040	0.312±0.054	3.274±0.154	0.820±0.007
SGS	0.150±0.021	0.389±0.018	3.064±0.052	0.860±0.018

(CER ± std: 0.119±0.040; UTMOS 3.274±0.154), while SGS demonstrates stronger speaker preservation (0.860 ± 0.018 vs 0.820 ± 0.007). The standard deviations indicate that SGS is more consistent across parameter conditions (lower CER and WER std) while BAS is more sensitive to parameter choices — consistent with the high-variance failure mode observed at BAS exag=0.7, cfg=0.5.

These complementary speaker profiles suggest that the fine-tuned T3 model generalises differently to different voice types: BAS produces more legible text reproduction, while SGS more faithfully preserves voice identity. This speaker-dependent behaviour may reflect differences in the speakers' acoustic characteristics relative to the training set speakers, or in the saliency of their voice embeddings to the frozen S3Gen generator.

F. Parameter Interaction Analysis

The ablation results across the exaggeration and CFG axes reveal structured main effects that provide guidance for practitioners. Table V summarises the mean performance for each parameter level, averaged across the other dimension and across both speakers.

TABLE V
MAIN EFFECTS OF EXAGGERATION AND CFG ON EACH METRIC (AVERAGED OVER THE OTHER FACTOR AND BOTH SPEAKERS).

Factor	Level	CER↓	WER↓	UTMOS↑	Sim.↑
Exag	0.3	0.125	0.326	3.228	0.830
Exag	0.7	0.145	0.377	3.110	0.850
CFG	0.3	0.120	0.334	3.149	0.835
CFG	0.5	0.149	0.368	3.189	0.845

The exaggeration parameter has the strongest effect on intelligibility: increasing exaggeration from 0.3 to 0.7 raises mean CER by ≈ 1.6 pp and WER by ≈ 5.1 pp, while reducing UTMOS by ≈ 0.12 points. In contrast, speaker similarity

increases with exaggeration ($+0.02$), indicating the parameter simultaneously trades intelligibility for speaker fidelity. The CFG parameter exhibits a qualitatively similar but weaker pattern: higher CFG marginally degrades CER/WER while improving speaker similarity. Critically, the two parameters are not fully collinear in their effects — the interaction between high exaggeration and high CFG is associated with the anomalously high CER variance observed for BAS (std 0.49), suggesting a synergistic instability under combined strong conditioning.

For Romanian TTS applications prioritising intelligibility (e.g., voice assistants, accessibility tools), we recommend exag=0.3, cfg=0.5, which achieves the best CER and WER. For applications prioritising voice identity fidelity (e.g., voice cloning, character synthesis), exag=0.7, cfg=0.3 or cfg=0.5 are preferable, with cfg=0.3 providing better stability.

G. Comparison with Prior Work

Table VI contextualises our results against the two closest comparable systems.

TABLE VI
COMPARISON WITH ROMANIAN TTS PRIOR WORK. (–) DENOTES CAPABILITY NOT SUPPORTED.

System	WER (%)↓	Spk. Sim.↑	Zero-shot
MMS-TTS-RON [8]	5.77	–	No
F5-TTS-RO (adapter) [7]	5.27	0.90	Yes
F5-TTS-RO (full FT) [7]	3.62	0.79	Yes
Ours (BAS, best)	**25.9**	**0.816**	**Yes**
Ours (SGS, best)	**37.2**	**0.875**	**Yes**

Our WER is higher than both prior systems, but direct comparison requires caution. F5-TTS-RO was evaluated on Common Voice Romanian, which comprises shorter, crowdsourced utterances; our evaluation uses SWARA holdout utterances of greater length and linguistic complexity. Furthermore, different ASR transcription systems were used. Speaker similarity is broadly comparable: our best condition achieves 0.875 vs the F5-TTS-RO adapter's 0.90, and substantially exceeds F5-TTS-RO full fine-tuning (0.79).

The key architectural distinction is that our approach requires *no model changes whatsoever*: no adapters, no new modules, no vocabulary extension. Only a 30-line text preprocessor and conservative fine-tuning of the existing T3 component are needed — a substantially lower implementation barrier than adapter-based methods. The higher WER also reflects limited training scale ($\sim$21 hours, 39 epochs) and suggests that further training would close the gap with prior work.

V. CONCLUSIONS AND FUTURE WORK

We have presented the first adaptation of the Chatterbox TTS architecture to a new language, demonstrating that phoneme-level text preprocessing enables stable fine-tuning of a BPE-tokenised Llama-based flow-matching model on Romanian without any modification to the model architecture or vocabulary. The central finding is that mean-initialised vocabulary extension causes posterior collapse — a failure that is language-agnostic and reproducible — while phoneme mapping to existing tokens fully avoids this problem by preserving the discriminative quality of pretrained embeddings. Systematic ablation over exaggeration and CFG inference parameters reveals a consistent intelligibility–speaker-fidelity trade-off that practitioners can exploit according to their application requirements: low exaggeration (0.3) and moderate CFG (0.5) maximise intelligibility (CER 8.7%, WER 25.9%), while higher exaggeration (0.7) improves voice-cloning fidelity (cosine similarity ≥ 0.87). All evaluated configurations produce intelligible Romanian speech with speaker similarity above 0.80 for unseen holdout speakers.

Several directions remain for future work. First, the training loss plateau after step 10,000 suggests continued improvement from lower learning rates and extended training duration. Second, integrating reinforcement learning with a Romanian ASR reward signal — following the Align2Speak [9] and F5R-TTS frameworks — could directly optimise intelligibility beyond what supervised fine-tuning alone achieves. Third, adapter-based hybrid approaches (following F5-TTS-RO [7]) could potentially combine the fidelity preservation of frozen modules with the phoneme-mapping strategy demonstrated here. Finally, subjective evaluation with native Romanian listeners and extension to spontaneous and expressive speech corpora remain important steps toward a production-grade Romanian TTS system.

REFERENCES

[1] C. Wang, S. Chen, Y. Wu, Z. Zhang, L. Zhou, S. Liu, Z. Chen, Y. Liu, H. Wang, J. Li, L. He, S. Zhao, and F. Wei, "Neural Codec Language Models are Zero-Shot Text to Speech Synthesizers," *arXiv preprint arXiv:2301.02111*, 2023.

[2] P. Anastassiou, J. Chen, J. Deng, Z. Gao, Y. Gu, Y. He, *et al.*, "Seed-TTS: A Family of High-Quality Versatile Speech Generation Models," *arXiv preprint arXiv:2406.02430*, 2024.

[3] Y. Chen, Z. Ye, J. Li, Z. Huang, J. Yang, S. Li, T. Bian, B. Wang, L. Wu, Z. Zhao, W. Chen, and B. Tian, "F5-TTS: A Fairytaler that Fakes Fluent and Faithful Speech with Flow Matching," in *Proc. ACL*, pp. 5904–5918, 2025.

[4] Z. Du, J. Chen, X. Guo, K. Chen, X. Zheng, Y. Zheng, *et al.*, "CosyVoice: A Scalable Multilingual Zero-shot Text-to-speech Synthesizer based on Supervised Semantic Tokens," *arXiv preprint arXiv:2407.05407*, 2024.

[5] Resemble AI, "Chatterbox: State-of-the-Art Open Source TTS," GitHub, 2025. [Online]. Available: https://github.com/resemble-ai/chatterbox

[6] R. Răgman and A. Stan, "Efficient Training Strategies for Natural Sounding Speech Synthesis in Romanian," *arXiv preprint arXiv:2410.06787*, 2024.

[7] A. Chivereanu and T. Boros, "Extending F5-TTS to Romanian TTS via Lightweight Input Adaptation," *arXiv preprint arXiv:2512.12297*, 2025.

[8] V. Pratap, A. Tjandra, B. Shi, P. Tomasello, A. Babu, S. Kundu, *et al.*, "Scaling Speech Technology to 1,000+ Languages," *J. Mach. Learn. Res.*, vol. 25, pp. 1–52, 2024.

[9] J. Chodroff, T. Mahajan, V. Pliakas, L. Saul, and A. Bhatt, "Align2Speak: Improving TTS for Low Resource Languages via ASR Preference Alignment," *arXiv preprint arXiv:2509.21718*, 2025.

[10] A. Stan, J. Yamagishi, S. King, and M. Aylett, "The SWARA Speech Corpus: A Large Parallel Romanian Read Speech Dataset," in *Proc. SPED*, 2017.

[11] T. Saeki, D. Xin, W. Nakata, T. Koriyama, S. Takamichi, and H. Saruwatari, "UTMOS: UTokyo-SaruLab System for VoiceMOS Challenge 2022," in *Proc. Interspeech*, pp. 4521–4525, 2022.

[12] A. Pîrlogeanu, B. Sas, V. Rachiţă, T.-M. Micu, and M. Dascalu, "Open Source State-Of-the-Art Solution for Romanian Speech Recognition," *arXiv preprint arXiv:2511.03361*, 2025.